

Beyond Missing Modalities: Hypergraph Guided Diffusion for Uncertainty-Aware Multimodal Emotion Recognition

Xihang Qiu^{1,2}, Yuhao Fang¹, Qing Zhou¹, Bin Zhai^{1,2}, Jialong Hong^{1,2},
 Wanpeng Zhang^{1,2}, Yao Lu¹, Ye Zhang^{1,2}, Chun Li^{1*}

¹ Shenzhen MSU-BIT University; ² Beijing Institute of Technology

qiuxh@bit.edu.cn; lichun2020@smbu.edu.cn

Abstract

*Multimodal Emotion Recognition in Conversations (MERC) aims to understand emotions expressed in each utterance by effectively integrating audio, text, and visual modalities. However, in real-world scenarios, unavoidable missing modalities often degrade multimodal interpretation performance. To address this, we propose **Hypergraph Diffusion and Evidence Fusion based Emotion Recognition (HyperEF)**, a novel framework designed to mitigate challenges arising from incomplete modalities in MERC. Specifically, to mitigate performance degradation caused by modality absence, we propose **Masked Hypergraph Attention (MHGAT)** conditioned diffusion model to recover latent features of missing modalities in the latent space. To ensure semantic consistency between recovered and available modalities within the same utterance, we introduce **MHGAT** that captures high-order semantic information from available modalities to guide the diffusion model's denoising process. Furthermore, to disentangle and model the complex uncertainties inherent in MERC, we propose **Dual Channel Evidence Fusion (DCEF)**, which estimates uncertainty at both feature source level and discriminative level, thereby achieving adaptive evidence fusion. Extensive comparative experiments and interpretability demonstrate the superior performance of our model in emotion recognition, as well as the contribution of each module within the model.*

1. Introduction

Multimodal Emotion Recognition in Conversations (MERC) is defined as the task of identifying emotional states by integrating signals from multiple modalities [21, 34, 42]. Compared to single-modal approaches, multimodal methods provide richer emotional understanding by leveraging complementary information across

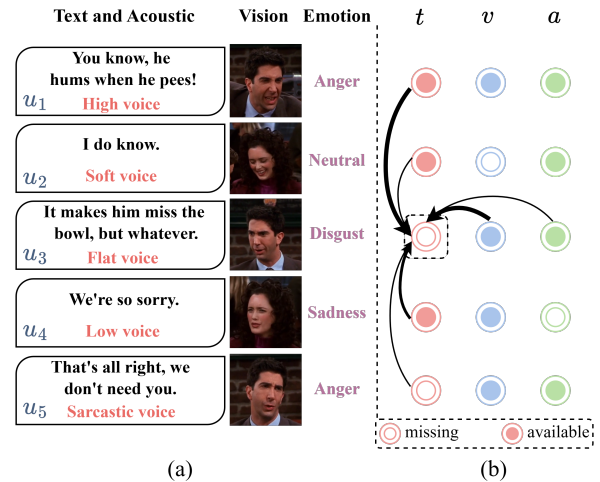


Figure 1. An example of multimodal dialogue (a) and the semantic inference process of text utterance u_3 (b).

modalities [13, 25, 27]. Such capabilities are essential in real-world applications including human-computer interaction [5, 40, 44] and mental health monitoring [6, 17], where accurate emotion perception significantly enhances user experience and system effectiveness.

However, modality-missing issues inevitably occur in real-world scenarios due to sensor failures or transmission errors [8, 23, 43, 47]. A common solution is modality recovery, which is categorized into two paradigms: **fusion-space completion** and **modality-space completion**. Fusion-space completion operates in the fused representation space but often suffers from weak interpretability and limited recovery quality [18]. Modality-space completion aims to reconstruct the latent features of missing modalities directly [37, 38]. A key challenge in this approach is the maintenance of semantic consistency between recovered and original modalities at the utterance level. This requires effective modeling of complex multivariate dependencies across dialogues, where emotional dependencies are often shaped by high-order information spanning both modality

*Corresponding authors.

and contextual dimensions [11, 17].

Figure 1(a) illustrates a dialogue between two speakers, and Figure 1(b) depicts the semantic inference process for the missing text modality of utterance u_3 . For example, at the modality level, the text and acoustic modalities of u_3 (ambiguous text and flat tone) present unclear semantic information; however, when combined with the visual modality, the emotion of disgust is suggested. At the context level, the implicit emotion in u_3 is closely related to utterances u_1 , u_4 , and u_5 . The apologetic response in u_4 and the emotional inertia of anger from the same speaker in u_1 and u_5 indicate that u_3 conveys a negative emotion associated with anger. Therefore, the relationships between utterances are complex and dynamic, involving high-order dependencies across both modalities and contextual dimensions. Moreover, as shown in Figure 1(b), during semantic inference of the u_3 text node, connected nodes contribute with non-uniform attention weights. At the modality level, the visual modality contributes more significantly than the acoustic modality; at the context level, contributions from u_1 , u_4 , and u_5 decrease sequentially, consistent with the previous analysis. Thus, designing a semantic capture network capable of selectively modeling high-order multivariate relationships in dialogues is critical for effective MERC [32], especially under incomplete modality conditions.

Existing multimodal methods typically treat all modalities equally or assign fixed weights to each modality [25, 41]. However, in practical scenarios, modalities provide signals of varying informativeness [16]. This issue is particularly critical in MERC with incomplete modalities [19], as features recovered by generative models have different confidence levels than available features. Additionally, discriminative confidence of each modality varies significantly across samples, often leading to ambiguous predictions. Therefore, an adaptive modality fusion approach is required to address the multidimensional uncertainty inherent in such scenarios.

To address these challenges, a novel framework named **Hypergraph Diffusion and Evidence Fusion based Emotion Recognition (HyperEF)**¹ is proposed. A **Masked Hypergraph Attention (MHGAT) Conditioned Diffusion** mechanism is introduced to recover missing modalities while minimizing semantic ambiguity between recovered and available modalities at the utterance level. Specifically, an MHGAT network is designed to capture high-order multivariate information from incomplete conversation data, which conditions the feature recovery process. Furthermore, to model uncertainties from both the feature source and discriminative layers, a **Dual Channel Evidence Fusion (DCEF)** mechanism is proposed to integrate multimodal data at the evidence level. By jointly captur-

ing source-layer uncertainty and discriminative ambiguity, the approach mitigates modality conflicts and strengthens evidence supporting the correct emotion category, thereby improving overall model performance.

In summary, the main contributions are as follows:

- **MHGAT Conditioned Diffusion:** A masked hypergraph attention network is introduced to capture high-order multimodal, contextual, and incomplete relations, guiding a diffusion model to achieve semantically consistent recovery of missing features.
- **Dual Channel Evidence Fusion:** Uncertainties from both the feature source and discriminative layers are disentangled and quantified to enable adaptive multimodal evidence fusion, effectively mitigating modality conflicts and enhancing recognition performance under incomplete modality conditions.
- Comprehensive comparative experiments, interpretability analyses, and ablation studies are conducted to demonstrate the effectiveness of the proposed model.

2. Related Work

2.1. Graph Neural Networks in MERC

Graph Neural Networks (GNNs) have been widely utilized to model data relationships, thus providing novel approaches for multimodal emotion recognition in conversations (MERC). Ai et al. [1] proposed a Fourier graph operator to extract long-range high- and low-frequency signals, effectively addressing the over-smoothing and low-pass filtering problems inherent to GNNs. Fu et al. [7] employed Relational Graph Neural Networks (RGNNs) to separately capture speaker-level and temporal information within dialogues. Despite their effectiveness, these methods are limited by the use of traditional pairwise relational structures, which hinder the modeling of complex multivariate interactions in multimodal conversations. To overcome this limitation, a hypergraph-based network was introduced by Chen et al. [3], enabling exploration of multivariate relationships and emotional dynamics through multi-frequency signal evaluation. Similarly, Chen and Shi [4] designed two sub-hypergraph networks to investigate consistent emotions in monologues and emotional shifts, respectively. However, these hypergraph-based approaches do not consider the varying contributions of different nodes and hyperedges. The proposed MHGAT model in this work is designed to address this gap by adaptively weighting the influence of nodes and hyperedges.

2.2. Evidence Theory

Evidence theory, commonly referred to as Dempster–Shafer Theory (DST) [31], provides a flexible framework for uncertainty modeling based on subjective belief functions. Unlike Bayesian approaches, which rely on parameter prob-

¹Code Repository: <https://github.com/wdqp/HyperDiff.git>.

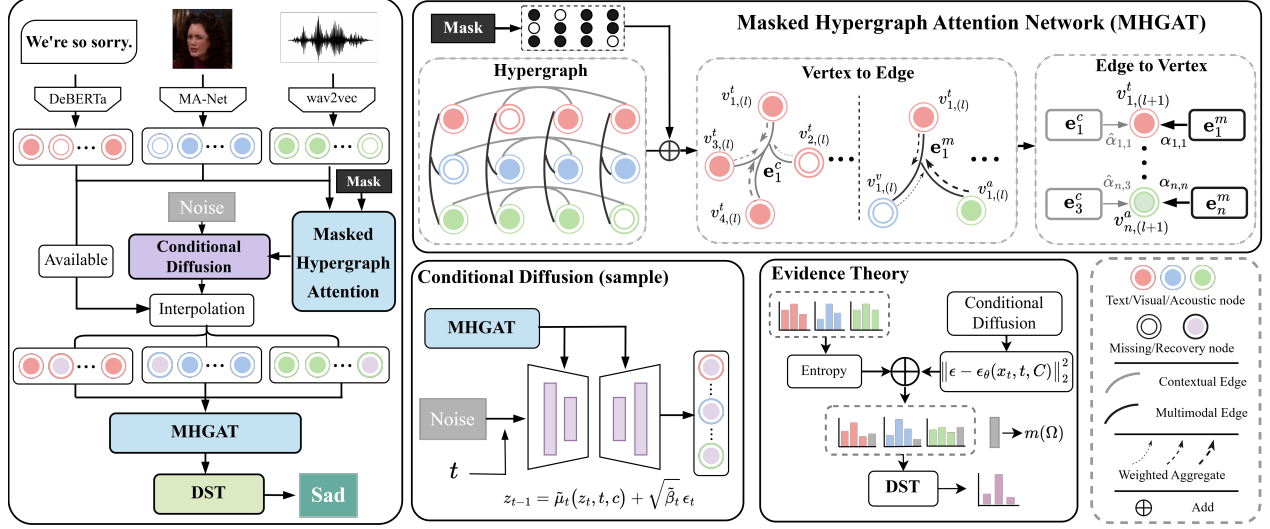


Figure 2. The detailed structure of the HyperEF. The leftmost box shows the overall framework, while the right side presents the details of each module. Mathematical symbols in the illustration are in line with the formulas in paper text.

ability distributions, DST assigns belief masses to subsets of possible outcomes and combines multiple sources of evidence through rules such as Dempster’s combination rule. Traditional DST-based uncertainty modeling primarily focuses on vacuity, representing the absence of supporting evidence. More recent studies have extended this framework to incorporate dissonance—arising from conflicting evidence—and consonance, which reflects the agreement among evidence on overlapping subsets [15]. These extended uncertainty measures provide a richer characterization of uncertainty and have been effectively applied in multimodal deep learning to improve robustness in scenarios with unreliable or missing modalities.

3. Methodology

In this section, we provide a detailed introduction to each component of the proposed HyperEF, as depicted in Fig 2.

3.1. MHGAT-Conditioned Diffusion

Our proposed MHGAT-Conditioned Diffusion aims to explicitly recover the latent features of missing modalities while ensuring semantic consistency between the recovered features and the original available features at the utterance level. We denote a dialogue with a sequence of N utterances as $U = \{(u_1, p_1), (u_2, p_2), \dots, (u_N, p_N)\}$, where each utterance u_i spoken by speaker p_i consists of three modalities. We use $u_i = \{u_i^t, u_i^v, u_i^a\}$ to indicate unimodal utterances from textual (t), visual (v) and acoustic (a) modalities. Considering the random modality missingness that occurs in real-world scenarios, we define a corresponding mask for each modality of every utterance: $\mathcal{M} = \{M_1, M_2, \dots, M_N\}$, $M_i = \{M_i^t, M_i^v, M_i^a\}$, where

1 means available modality and 0 means missing modality. Define the set of emotion labels $Y = [y_1, y_2, \dots, y_c]$, where c denotes the number of emotional categories.

3.1.1. Masked Hypergraph Attention Network

To alleviate semantic confusion between the recovered and original features, we propose the MHGAT to adaptively capture high-order dependencies in multimodal dialogue data, where the mask mechanism guides the model to differentiate between available and missing modalities.

Given a dialogue U , we construct a hypergraph $\mathcal{H} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{v_1, \dots, v_n\}$ ($|\mathcal{V}| = 3N$) represents a set of all unimodal utterances, and hyperedges $\mathcal{E} = \{e_1^c, e_2^c, e_3^c, e_1^m, \dots, e_N^m\}$ ($|\mathcal{E}| = N + 3$) contains contextual edges (e_i^c) and multimodal edges (e_i^m). For convenience, we define $n = |\mathcal{V}|, m = |\mathcal{E}|$. Note that for any hyperedge e , it can connect more than two nodes. The clear structure is shown in the Fig 2. When constructing node v_i , we preserve the hyperedge relation and positions of missing nodes, only setting their values to zero. Then we introduce the corresponding embedded mask to all nodes: $v_i = v_i + \text{embedding}(M_i)$.

Traditional hypergraph networks rely on a fixed incidence matrix $A \in \mathbb{R}^{n \times m}$, where $A_{i,j} = 1$ if v_i is incident with e_j , else $A_{i,j} = 0$. However, the inherent static of H limits the model’s ability to adaptively prioritize more informative relationships during the aggregation process. Our MHGAT addresses this issue via an attention mechanism. MHGAT learns node representations with two different aggregation functions:

$$\begin{aligned} e_j^{l+1} &= AGG_{v \rightarrow e}^l(e_j^l, \{v_i^l \mid v_i \in \mathcal{V}_j\}), \\ v_i^{l+1} &= AGG_{e \rightarrow v}^l(v_i^l, \{e_j^{l+1} \mid e_j \in \mathcal{E}_i\}), \end{aligned} \quad (1)$$

where $\mathcal{V}_j$ denotes the set of nodes connected to edge e_j , while $\mathcal{E}_i$ denotes the set of hyperedges connected to node v_i . we first aggregate the nodes to get the hyperedges in the next layer through $AGG_{v \rightarrow e}(\cdot)$, then aggregate the new hyperedges to update the nodes through $AGG_{e \rightarrow v}(\cdot)$. Both functions are implemented through attention mechanism.

Node to Hyperedge: Given any hyperedge e_j^l , our MHGAT first aggregates information from its connected nodes $v_i^l \in \mathcal{V}_j$ to obtain the hyperedge representation at the next layer. Since not all nodes contribute equally to the hyperedge, we employ an attention mechanism to emphasize the more important nodes while down-weighting those ambiguous ones. The aggregate function is:

$$e_j^{l+1} = \sigma\left(\sum_{v_i \in \mathcal{V}_j} \alpha_{ji} W_1 v_i^l\right), \quad (2)$$

where σ is a nonlinearity such as ReLU, W_1 is a trainable weight matrix, and α_{ji} is the attention score between e_j and v_i , which can be calculated by:

$$\alpha_{ji} = \frac{\exp\left(\text{LeakyReLU}\left(\mathbf{a}_1^\top [W_1 v_i^l \parallel W_2 e_j^l]\right)\right)}{\sum_{v_k \in \mathcal{V}_j} \exp\left(\text{LeakyReLU}\left(\mathbf{a}_1^\top [W_1 v_k^l \parallel W_2 e_j^l]\right)\right)}, \quad (3)$$

where $\mathbf{a}_1^\top$ is a trainable weight vector, W_2 is a trainable weight matrix, and $\parallel$ is the concatenation operation. Owing to the absence of intrinsic features for hyperedges, we represent each hyperedge by a learnable embedding vector, i.e., $e_j^{(0)} \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$.

Hyperedge to Node: With all the hyperedge representations, we aggregate the ones $e_j \in \mathcal{E}_i$ connected to v_i to update the node. The aggregate functions can be described as:

$$v_i^{l+1} = \sigma\left(\sum_{e_j \in \mathcal{E}_i} \beta_{ij} W_2 e_j^{l+1}\right), \quad (4)$$

where β_{ij} is the attention score of e_j on v_i .

$$\beta_{ij} = \frac{\exp\left(\text{LeakyReLU}\left(\mathbf{a}_2^\top [W_2 e_j^{l+1} \parallel W_1 v_i^l]\right)\right)}{\sum_{e_k \in \mathcal{E}_i} \exp\left(\text{LeakyReLU}\left(\mathbf{a}_2^\top [W_2 e_k^{l+1} \parallel W_1 v_i^l]\right)\right)}, \quad (5)$$

where $\mathbf{a}_2^\top$ is another weight vector.

The proposed MHGAT not only captures multivariate relationships in multimodal dialogues and highlights key information at different granularities during node representation learning, but also learns to distinguish missing modalities based on the mask mechanism.

3.1.2. Conditional Diffusion

Inspired by Classifier-Free Guidance, we concatenate the updated node representations from MHGAT along the modality dimension and use them as the condition $C \in \mathbb{R}^{n \times 3d}$ in the conditional diffusion model. This condition is applied in both the training and sampling stages to guide

the diffusion model in generating recovered features that are semantically consistent with the original features.

Training process: We implement the conditional noise prediction network with a U-Net architecture, where cross-attention layers are used to integrate the intermediate feature of the U-Net with the conditioning signal C :

$$F^{(l+1)} = \text{Concat}(F^{(l)}, \text{CrossAttn}(F^{(l)}, \text{Linear}(C))), \quad (6)$$

where $F^{(l)}$ is the input of the l^{th} layer of U-Net, $\text{Linear}(\cdot)$ projects C to the same size of $F^{(l)}$. To amplify the condition, we randomly omit the condition with a fixed probability p during training to enable unconditional predictions. The objective function is:

$$\mathcal{L} = \mathbb{E}_{\mathbf{x}_0, C, t, \epsilon \sim \mathcal{N}(0, \mathbf{I})} \left[\|\epsilon_\theta(\mathbf{x}_t, C, p) - \epsilon\|^2 \right] \quad (7)$$

Sampling process: During inference we merge the conditional and unconditional predicted noise with a weighted combination:

$$\hat{\epsilon}_\theta(\mathbf{x}_t, C) = \epsilon_\theta(\mathbf{x}_t, \emptyset) + w \cdot (\epsilon_\theta(\mathbf{x}_t, C) - \epsilon_\theta(\mathbf{x}_t, \emptyset)) \quad (8)$$

where $\emptyset$ means no condition, and w is used to balance the model's weighting between the conditional signal and the unconditional noise. During the reverse denoising process, we guide the noise prediction with the conditional embedding and define the reverse transition distribution as:

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t, C) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\mu}_t(\mathbf{x}_t, t, C), \sigma_t^2 \mathbf{I}), \quad (9)$$

where $\tilde{\mu}_t(\mathbf{x}_t, t, C)$ is calculated as:

$$\tilde{\mu}_t(\mathbf{x}_t, t, C) = \frac{1}{\sqrt{\eta_t}} \left(\mathbf{x}_t - \frac{1 - \eta_t}{\sqrt{1 - \eta_t}} \hat{\epsilon}_\theta(\mathbf{x}_t, t, C) \right), \quad (10)$$

where $\eta_t = 1 - \beta_t$ is the retention rate of step t , and $\bar{\eta}_t = \prod_{s=1}^t \eta_s$. The final sampling process is:

$$\mathbf{x}_{t-1} = \tilde{\mu}_t(\mathbf{x}_t, t, C) + \sigma_t \cdot z, \quad z \sim \mathcal{N}(0, \mathbf{I}). \quad (11)$$

3.2. Dual Channel Evidence Fusion

The traditional DST computes the basic probability assignment (BPA) for the universal set based on vacuity. However, vacuity reflects uncertainty only from the perspective of the evidence sum, which fails to capture the complexity and multi-dimensional entanglement of uncertainty in multimodal scenarios. We propose DCEF, which models the complex uncertainty by disentangling uncertainty at both the feature source level and the discriminative level.

3.2.1. Aggregation Strategy with DCEF

Evidence theory associates the parameter α of the Dirichlet distribution $\text{Dir}(\mathcal{P}|\alpha)$ with belief distributions, where $\mathcal{P}$

represents the simplex that indicates class assignment probabilities. Evidence theory suggests that by simply replacing the final softmax layer of a conventional neural network with a ReLU activation layer, it can be transformed into an evidential neural network, where the non-negative output of the ReLU layer is defined as evidence.

Since the feature recovery in MHGAT introduces additional sources of uncertainty, conventional vacuity-based estimators fail to capture such complex uncertainty. We therefore disentangle the BPA of the universal set Ω into sufficient yet minimally redundant components, enabling a more comprehensive estimation of uncertainty. Existing methods typically decompose uncertainty into epistemic and aleatoric components [24, 26]. However, Matias et al. [35] shows that these two factors are not strictly independent and often interact in practice. To obtain a decomposition tailored to MHGAT-Diffusion, we first anchor one axis of uncertainty at the **feature source level**. we estimate the uncertainty of the recovered features based on the mean squared error (MSE) between the predicted noise and the actual noise at the last timestep during the sampling process of the conditional diffusion model:

$$\begin{aligned} m_s(\Omega) &= a \cdot \text{Sigmoid}(b \cdot (u - \bar{u})), \\ u &= \|\epsilon - \epsilon_\theta(\mathbf{x}_t, T, C)\|_2^2, \end{aligned} \quad (12)$$

where $\bar{u}$ is the mean of u , T means the last timestep, a and b are scaling factors. Considering $0 \leq m(\Omega) < 1$, we use $\bar{u}$ and the Sigmoid function to map u near 0.5, and then scale and fine-tune it to an appropriate range with a and b . Besides, $m_s(\Omega)$ for original features are set to 0.

To ensure the uncertainty decomposition satisfies sufficiency and low redundancy, we evaluate four additional uncertainty measures: *vacuity*, *dissonance*, *consonance* and *entropy*. By pairing each with the anchored measure, we identify **discriminative level** as the optimal second axis. Detailed experiments are reported in Sec. 4.8. At the **discriminative level**, we employ normalized cross-entropy to estimate uncertainty:

$$m_d(\Omega) = \frac{-\sum_{k=1}^K P_k \log(P_k)}{\log K}, \quad (13)$$

since the higher entropy typically leads to decision ambiguity, reflecting greater uncertainty.

Here, we present a method for estimating the BPA of the universal set Ω through a dual-channel approach in a unimodal setting. Given the i^{th} sample in modality $m \in \{t, v, a\}$, let $e_i^m = \{e_{i1}^m, e_{i2}^m, \dots, e_{iK}^m\}$ represent the evidence vector for a K -class classification problem. The corresponding Dirichlet distribution parameter can be denoted as $\alpha_i^m = e_i^m + 1 = \{\alpha_{i1}^m, \dots, \alpha_{iK}^m\}$. Then, we can estimate each class's probability by $P_{ik}^m = \alpha_{ik}^m / S_i^m$, where $S_i^m = \sum_{k=1}^K \alpha_{ik}^m$ is the Dirichlet strength. In order to simplify the description while maintaining generality, we will

omit m and i representing modalities and samples. Then we can get the BPA of the universal set and each class:

$$\begin{aligned} m(\Omega) &= \gamma \cdot m_d(\Omega) + m_s(\Omega), \\ m(k) &= (1 - m(\Omega))P_k, \end{aligned} \quad (14)$$

where γ is a scaling factor. According to DST theory, we iteratively fuse pairwise and obtain the final fused probability vector using the following formula:

$$(m_1 \oplus m_2)(A) = \frac{\sum_{B \cap C = A} m_1(B)m_2(C)}{1 - \sum_{B \cap C = \emptyset} m_1(B)m_2(C)}, \quad (15)$$

where m_1 and m_2 denote distinct sources of evidence, which correspond to the output probability distributions of different modalities in this work. A , B , and C denote arbitrary classes, and note that Ω contains all subsets.

3.2.2. Learning From Uncertainties Estimation

Inspired by [14], we discuss three main types of uncertainty in the design of the DCEF and optimization objective function: *vacuity*, *dissonance* and *consonance*. Specifically, vacuity uncertainty is defined as:

$$Vac(\alpha) = \frac{K}{S}, \quad (16)$$

where $S = \sum_{k=1}^K \alpha_k$. The dissonance uncertainty is defined as:

$$Dis(\alpha) = \sum_{k=1}^K \left(\frac{b_k \sum_{j \neq k} b_j \text{Bal}(b_j, b_k)}{\sum_{j \neq k} b_j} \right), \quad (17)$$

where $b_k = \frac{e_k}{S}$ and

$$\text{Bal}(b_j, b_k) = \begin{cases} 1 - \frac{|b_j - b_k|}{b_j + b_k} & \text{if } b_j b_k \neq 0, \\ 0 & \text{otherwise,} \end{cases} \quad (18)$$

and the consonance uncertainty is defined as:

$$Con(\alpha) = \frac{1}{b_s} \sum_{b=1}^{b_s} U_b, \quad (19)$$

$$U_b = \frac{1}{\binom{M}{2}} \sum_{1 \leq i < j \leq M} \left(1 - \frac{\sum_{k=1}^K \min(\alpha_{b,k}^{(i)}, \alpha_{b,k}^{(j)})}{\sum_{k=1}^K \max(\alpha_{b,k}^{(i)}, \alpha_{b,k}^{(j)}) + \varepsilon} \right), \quad (20)$$

where ε is a small constant to prevent division by zero, and i, j represent different modal

$Con(\alpha)$ is used to evaluate the degree of semantic confusion between modalities, which is caused by the differences in evidence between different modalities. $Dis(\alpha)$ is uncertainty caused by evidence conflict within one modality. The primary cause of $Vac(\alpha)$ is the lack of evidence, i.e., insufficient information.

Dataset	Method	Venue	Missing Rate							
			0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7
IEMOCAP4	MMIN [45]	ACL'21	74.9	71.8	69.4	66.3	63.3	60.5	57.5	55.4
	GCNET [18]	TPAMI'23	78.4	77.5	77.3	76.2	75.1	73.8	71.9	71.4
	IMDer [39]	NeurIPS'23	74.3	73.5	69.5	64.1	59.2	64.5	46.9	44.2
	CIF-MMIN [20]	TAFCC'24	79.3	77.9	77.4	77.3	77.1	<u>76.0</u>	74.7	73.6
	SDR-GNN [7]	KBS'25	<u>79.6</u>	<u>78.6</u>	<u>78.1</u>	<u>77.5</u>	<u>77.1</u>	<u>75.8</u>	<u>75.0</u>	<u>74.4</u>
	HyperEF (Ours)	—	82.9 (↑3.3)	82.1 (↑3.5)	79.1 (↑1.0)	80.3 (↑2.8)	77.6 (↑0.5)	78.2 (↑2.2)	77.7 (↑2.7)	77.0 (↑2.6)
IEMOCAP6	MMIN [45]	ACL'21	57.0	53.9	51.5	48.4	45.6	42.8	40.2	37.8
	GCNET [18]	TPAMI'23	58.6	58.5	57.6	57.1	56.1	54.4	53.6	53.5
	IMDer [39]	NeurIPS'23	57.9	52.9	53.5	46.0	49.9	39.6	31.1	28.2
	CIF-MMIN [20]	TAFCC'24	60.5	58.8	57.3	56.8	56.2	55.7	55.1	54.3
	SDR-GNN [7]	KBS'25	<u>61.2</u>	<u>60.9</u>	<u>59.8</u>	<u>59.5</u>	<u>59.2</u>	<u>57.4</u>	<u>55.5</u>	<u>55.3</u>
	HyperEF (Ours)	—	64.1 (↑2.9)	63.2 (↑2.3)	61.8 (↑2.0)	61.1 (↑1.6)	60.1 (↑0.9)	59.8 (↑2.4)	59.5 (↑4.0)	58.5 (↑3.3)
MELD	MMIN [45]	ACL'21	64.9	59.8	58.8	55.8	54.8	53.7	51.3	49.3
	GCNET [18]	TPAMI'23	65.5	63.3	62.2	59.7	58.3	57.2	55.2	53.6
	IMDer [39]	NeurIPS'23	<u>67.1</u>	<u>66.1</u>	62.6	61.4	<u>60.5</u>	<u>58.4</u>	<u>57.0</u>	<u>55.3</u>
	CIF-MMIN [20]	TAFCC'24	65.8	65.3	<u>62.7</u>	61.2	59.6	58.0	56.4	55.3
	SDR-GNN [7]	KBS'25	63.8	62.4	60.9	57.8	57.4	56.6	54.4	52.9
	HyperEF (Ours)	—	69.3 (↑2.2)	67.2 (↑1.1)	64.8 (↑2.1)	63.8 (↑2.4)	61.8 (↑1.3)	60.4 (↑2.0)	58.6 (↑1.6)	57.1 (↑1.8)

Table 1. Accuracy (%) comparison of methods on IEMOCAP4, IEMOCAP6, and MELD datasets under varying missing modality rates. We bold the highest accuracy and underline the second-best accuracy.

Since $Vac(\alpha)$ indiscriminately amplifies evidence for all classes, we introduce a Kullback-Leibler (KL) divergence regularizer to penalize the incorrect classes:

$$\mathcal{L}_{KL} = \arg \min_{\tilde{\alpha}_i} \sum_{i=1}^n \text{KL} [\text{Dir}(P_i | \tilde{\alpha}_i) \| \text{Dir}(P_i | \langle 1, \dots, 1 \rangle)], \quad (21)$$

where $\tilde{\alpha}_i = y_i + (1 - y_i) \odot \alpha_i$, y_i is the one-hot label of sample i , and $\text{Dir}(P_i | \langle 1, \dots, 1 \rangle)$ means the uniform Dirichlet distribution.

Since an excess of regularizers and hyperparameters can harm stability, we analyze the four candidates for their impact on accuracy, evidence, and output entropy, and find that Con and Dis have negligible impact, whereas Vac and $\mathcal{L}_{KL}$ provide complementary, effective constraints (detailed in Appendix Sec. 13). We then select Vac and $\mathcal{L}_{KL}$ as to form the final objective:

$$\mathcal{L} = \arg \min_{\alpha_i} \sum_{i=1}^n \mathcal{L}_{CE}(P_i, y_i) + \lambda_1 \mathcal{L}_{KL}(\alpha_i) + \lambda_2 Vac(\alpha_i), \quad (22)$$

where λ_1, λ_2 are balance factors whose optimal values are determined by Bayesian optimization.

4. Experiments

4.1. Datasets

To verify the effectiveness of HyperEF, we conduct experiments on two benchmark conversational datasets: IEMOCAP [2] and MELD [28]. IEMOCAP has 151 dialogues and 7433 utterances between 2 speakers. For a fair comparison, we employ two prevalent labeling methods, generating datasets with four classes [18] and six classes [7]. The

MELD dataset contains 1,432 multi-party conversational clips from the TV series Friends, encompassing 13,708 utterances and 304 unique speakers with seven emotion types.

4.2. Experimental Settings

In line with previous works [7, 18, 38], we test different models on a commonly used random missing protocol. We define the missing rate as $\kappa = 1 - \frac{\sum_{i=1}^N m_i}{N \times M}$, where m_i denotes the number of available modalities for the i^{th} sample, N indicates the total number of samples, and M is the number of modalities. We ensure that at least one modality is available for each sample, s.t. $m_i \geq 1$ and $\kappa \leq \frac{M-1}{M}$. Here we set M to 3, so we have $\kappa \in [0.0, 0.1, \dots, 0.7]$.

Due to different numbers of emotion categories and speakers between the two datasets, separate parameters and models are set. For IEMOCAP, we set the batch size of 8 dialogues, and use AdamW with a learning rate of 5×10^{-5} . For MELD, we set the batch size of 64 dialogues, and use AdamW with a learning rate of 1×10^{-4} . For the MHGAT Conditioned Diffusion, we set the layer number of MHGAT to 2, the timesteps of DDPM-based diffusion model to 300, and condition weight w to 7.5. For DCEF, we first perform a coarse grid search over a, b, γ , constraining $a \in [0.1, 0.2], b \in [3, 6], \gamma \in [0.1, 0.3]$, then apply Bayesian optimization for a fine-grained search over $\lambda_{1-2}, a, b, \gamma$. Finally, we set $\bar{u} = 1.3, b = 5, a = \frac{1}{7}$, and $\gamma = 0.2$. For the loss function, we set $\lambda_1 = 0.5, \lambda_2 = 0.8$. We perform five-fold cross-validation following [18] and report the mean values on the test set. All experiments were conducted on a NVIDIA L40S GPU with 48 GB memory. The model training has two stages: 1) pre-train the MHGAT conditioned diffusion model for 100 epochs; 2) the backbone network is trained for 50 epochs.

Method	Param	Times	MDLE		IEMOCAP4	
			0.3	0.5	0.3	0.5
MHGAT	1.60M	0.82s	62.4	59.6	78.7	77.1
Trans	6.83M	0.88s	62.5	59.4	78.1	76.2

Table 2. Performance and cost of MHGAT and Transformer encoder on MDLE and IEMOCAP4 at missing rates 0.3 and 0.5.

4.3. Baselines

4.3.1. Fusion-space Completion

GCNet [18] and SDR-GNN [7] leverage GNN to extract semantic information from multimodal dialogue, recover missing information at the fused feature level.

4.3.2. Modality-space Completion

CIF-MMIN [20] and MMIN [45] are both extensions of autoencoders. IMDer [39] uses a score-based diffusion model to explicitly restore the features of missing modalities.

4.4. Overall Results

Following previous studies [7, 18], we evaluate four class accuracy (IEMOCAP4) and six class accuracy (IEMCAP6) on IEMOCAP. For MELD, we report the seven class accuracy.

As shown in table 1, we compare our HyperEF approach with state-of-the-art (SOTA) MERC methods under unified environmental and dataset settings. It turns out that HyperEF has the best emotion recognition effect on all datasets and under all missing rates. Specifically, on the IEMOCAP4 dataset, HyperEF is 3.5% better than SDR-GNN under the missing rate of 0.1. On the IEMOCAP6 benchmark, under elevated missing rates, HyperEF yields substantially greater accuracy gains over baseline methods (4.0% under missing rate 0.6), owing to its MHGAT-based semantic feature reconstruction. Similarly, on the MELD dataset, HyperEF also achieves the best emotion recognition performance. Across the eight missing-rate settings on all datasets, all paired differences are in the same direction; hence the Wilcoxon signed-rank test attains the extreme statistic $W = 0$ with an exact two-sided $p = 0.0078$ for every baseline comparison.

4.5. MHGAT vs. Transformer encoder

Given that a Transformer encoder can, through self-attention, naturally learn complex relations among arbitrary elements, we compare MHGAT with a 2 layer Transformer baseline to validate the advantages of MHGAT. We attach an identical classification head to both networks and use accuracy as a proxy for their ability to capture semantic information. As shown in Table 2, MHGAT uses 23% of the parameters of the Transformer while achieving comparable or slightly higher accuracy on MELD and IEMOCAP. The

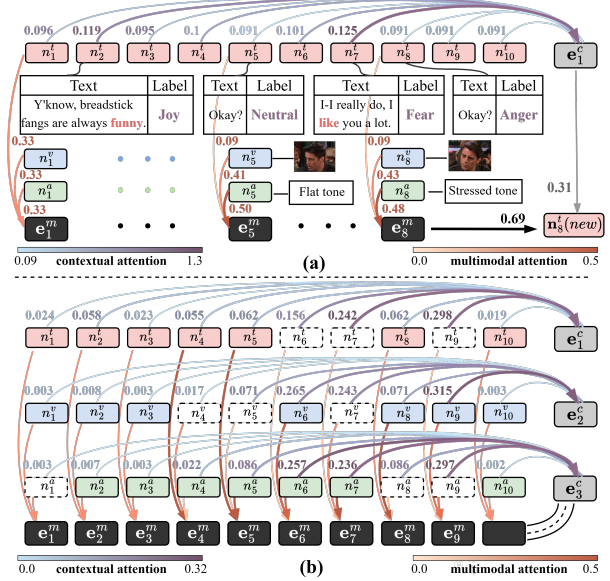


Figure 3. Aggregation process of MHGAT and visualization of attention assigned to each node and edge.

‘Times’ column reports the average per-epoch wall clock time to run two layers on IEMOCAP. The Transformer encoder configuration and analysis of time complexity are provided in the Appendix Sec. 10.

4.6. Attention visible

Fig 3 (a) and (b) depict the attention visualizations of ‘Node to Hyperedge’ and ‘Hyperedge to Node’ steps in MHGAT under full-modality and missing-modality (missing rate = 0.3) conditions respectively.

Specifically, Fig 3 (a) illustrates the ‘Node to Hyperedge’ for the textual modality and the ‘Hyperedge to Node’ for the eighth node. From the ‘Node to Hyperedge’ step, we can observe that in the t modality’s contextual dimension, the model assigns higher attention scores to utterances containing strong emotional words like ‘funny’ and ‘like’, while ambiguous expressions like ‘Okay’ receive lower scores. Furthermore, when the text modality provides insufficient semantic cues, the model adaptively shifts attention to the other modalities and, during the ‘Hyperedge to Node’, strengthens the weights of multimodal hyperedges to recover semantic information; see node 8 for an illustrative example.

In Fig 3 (b), it is evident that, after incorporating the missing modality mask information, all three modalities in the contextual dimension tend to assign higher attention scores to node 6, 7, and 9. While in the multimodal dimension, the model tends to allocate lower attention scores to the missing modalities. This demonstrates that the mask mechanism can capture information about whether a modality is missing.

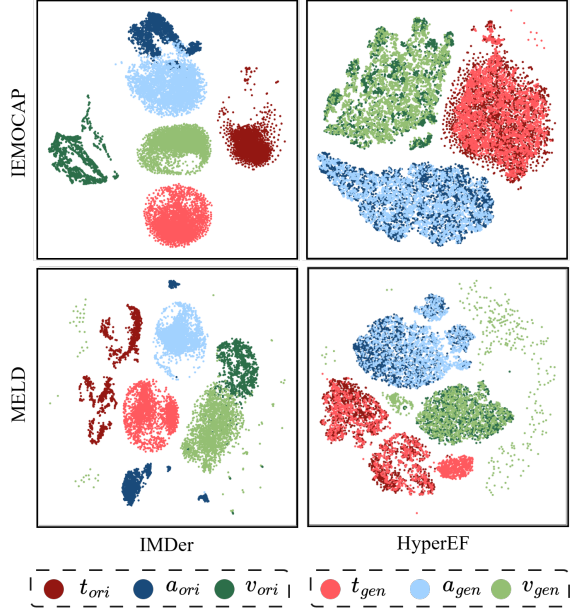


Figure 4. t-SNE visualization of recovered and original features from HyperEF and IMDer on two datasets.

Method	Metric	MELD			IEMOCAP		
		t	a	v	t	a	v
HyperEF	MSE	0.028	0.051	0.070	0.036	0.024	0.031
	MMD	0.017	0.024	0.001	0.016	0.012	0.021
IMDer	MSE	0.701	2.350	2.846	0.151	4.883	0.503
	MMD	0.487	0.365	0.227	0.388	0.265	0.425

Table 3. Recovery error (MSE) and distribution discrepancy (MMD) on MELD and IEMOCAP across all modalities ($t/a/v$).

4.7. Recovery performance

Fig 4 and Table 3 illustrate the feature recovery capabilities of HyperEF compared to IMDer. We projected both the recovered and original features into a two-dimensional space and visualized them using t-SNE plots. It is evident that our model achieves superior recovery performance compared to IMDer on both datasets. On the both datasets, across all three modalities, our model recovers features that exhibit highly similar data distributions to the original features.

To quantitatively assess the recovered features, we use Mean Squared Error (MSE) to measure per sample fidelity and squared Maximum Mean Discrepancy (MMD) with an RBF kernel to evaluate distribution level alignment. The results in Table 3 show that our method achieves the best recovery for every modality across both datasets.

We provide analysis of the time complexity introduced by the diffusion model in Appendix Sec. 14.

4.8. Evidence disentanglement

Table 4 details the procedure in Sec. 3.2.1 for identifying the optimal second disentanglement direction. We pair our

$m_s(\Omega)$	Con	Dis	Vac	$Entropy$	MELD	
					Acc.	F1
$baseline$					60.28	57.93
✓	✓	✗	✗	✗	60.91	56.81
✓	✗	✓	✗	✗	62.56	58.89
✓	✗	✗	✓	✗	59.87	55.90
✓	✗	✗	✗	✓	63.78	60.47

Table 4. Performance of different disentanglement strategies on MELD with a missing rate of 0.3

$\mathcal{L}_{KL}$	Vac	IEMOCAP4		MELD	
		Acc.	F1	Acc.	F1
✗	✗	76.85	76.71	55.06	43.35
✗	✓	48.75	38.55	62.92	59.76
✓	✗	77.60	77.42	50.40	33.78
✓	✓	80.31	79.77	63.78	61.47

Table 5. Ablation study on IEMOCAP4 and MELD with a 0.3 missing rate at different target function switches.

anchored $m_s(\Omega)$ with each of four additional uncertainty measures and evaluate on MELD with a 0.3 missing rate. The baseline is vacuity-based evidence fusion. The results indicate that the discriminative level uncertainty based on entropy is the optimal second disentanglement direction.

4.9. Ablation Study

To investigate the effectiveness of the target function, we sequentially removed the terms from the optimization objective and conducted experiments on the IEMOCAP4 and MELD datasets with a 0.3 missing rate. The baseline only uses $\mathcal{L}_{CE}(P_i, y_i)$. The results are presented in Table 5. On the IEMOCAP dataset, removing $\mathcal{L}_{KL}$ prevents the model from effectively penalizing incorrect categories, resulting in a significant decrease in accuracy. On the MELD dataset, removing Vac leads to insufficient evidence, which substantially reduce accuracy.

5. Conclusion

We propose a novel MERC model named HyperEF. The MHGAT-Diffusion explicitly recovers missing modality features by leveraging high-order relational information captured by MHGAT to guide the denoising process, thereby reducing semantic ambiguity between recovered and original features. The DCEF models uncertainty at both the discriminator and feature source levels, achieving a sufficient and minimally redundant disentanglement of uncertainty. Extensive comparisons validate our model, and targeted interpretability and ablations show the performance of our model.

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